



Mario Varona Bueno

Senior iOS Engineer

MSc. in Cybersecurity

MSc. in Business Intelligence and Big Data in Cyber-Secure Environments

About me

I enjoy building high-quality software that solves real-world problems.

I value effort, continuous learning, autonomy, and ownership to take projects end-to-end.

With a product engineer mindset, I bring a holistic perspective: strong technical experience in mobile, web, cybersecurity and Big Data, combined with experience leading teams and supporting people to help them shine through empathetic leadership.

I believe in collaborating with diverse profiles and in user-centered, iterative development to create technology that improves people's lives.

Education

Universities of Burgos, León and Valladolid (2021 - 2023)

Interuniversity Master in Business Intelligence and Big Data in Cyber-Secure Environments

Average grade: 3.67/4 GPA (9.17/10)

Master thesis: Online recommendation system for Spanish municipalities based on user preferences and similarity across municipalities

Master thesis' grade: 3.92/4 GPA (9.8/10)

Complutense University of Madrid (2021 - 2022)

Master in Cybersecurity

Average grade: 3.64/4 GPA (9.1/10)

Master thesis: Design of a cybersecurity program for a startup

Munich University of Applied Sciences (2020)

Computer Networking and Secure Network Management

Julius-Maximilians-Universität Würzburg (2019 - 2020)

One-year stay on Würzburg, Bayern, Germany

University of Salamanca (2017 - 2021)

Computer Science Degree

Average high school scientific baccalaureate grade: 3.72/4 GPA (9.29/10)

Average degree grade: 3.33/4 GPA (8.32/10)

Member of the ACM USAL chapter since 2018

Bachelor thesis: Integral system to perform on-board payments on public transport with and without NFC, platform to manage it, environment to test it, app to calculate avoided CO2 and platform to notify user

Bachelor thesis' grade: A+ (10/10)

Experience



iOS Engineer (2026 - ...)

Qustodio by Qoria — <https://www.qustodio.com>

Development and maintenance of parental control features, including network traffic monitoring and filtering, enforcement of access policies or location and device management, focusing on security and user experience for minors.

Technologies and techniques used: Swift, SwiftUI, Network Extension, VPN & Packet Tunnel, Mobile Device Management (MDM), TLS inspection, SOLID principles, Clean Code, Clean Architecture, Continuous Integration.

Skills

iOS Development

- Swift
- SwiftUI
- Objective-C
- UIKit
- TDD
- Clean Architecture

Android Development

- Java
- Kotlin

Big Data

- Python
- Pandas

Web Development

- HTML
- CSS
- JavaScript
- Vue.js
- jQuery
- Flask, FastAPI
- PHP, MySQL, NoSQL (MongoDB, Firestore)

DevOps / CI/CD

- Google Cloud Platform
- GitHub Actions
- Fastlane

Agile Methodologies

- Scrum
- Kanban

Communication

- Teamwork
- Empathetic leadership

 **Spanish:** Native

 **English:**

- C1, OLS, 2019
- B2, Cambridge FCE, 2016

 **German:**

- B2, Julius-Maximilians-Universität Würzburg, 2020

Contact

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 [MarioVarona](https://github.com/MarioVarona)

 [MarioVarona](https://www.linkedin.com/in/MarioVarona)



Senior iOS Engineer (2025 - 2026)

Badger Maps, Inc. — <https://www.badgermapping.com>

Development of features for user acquisition and retention with a Product Engineering approach focused on delivering value to the user incrementally. Creation and improvement of internal tools and pursuit of process optimization.

Technologies and techniques used: Swift, SwiftUI, Objective-C, UIKit, TDD, BDD, DDD, SOLID principles, Clean Code, Clean Architecture, Continuous Integration, Continuous Delivery.



Interim Engineer Manager (2024 - 2024)

Badger Maps, Inc. — <https://www.badgermapping.com>

Accompaniment of 5 direct reports through biweekly 1:1, fostering a stronger writing and feedback culture, conducting the first DevEx survey, encouraging greater engineering ownership, enhancing visibility into Engineering & Product metrics, assessing the current state of technical debt and monitoring alerts, or promoting continuous learning through courses and self-development initiatives.



iOS Engineer (2022 - 2025)

Badger Maps, Inc. — <https://www.badgermapping.com>

Seeking a significant qualitative leap in the iOS project, by including advanced mocking and testing tools, migrating critical code from Objective-C and UIKit to Swift and SwiftUI, detecting potential bottlenecks and providing proposals to reduce them.



Junior iOS Engineer (2022 - 2022)

Badger Maps, Inc. — <https://www.badgermapping.com>

Work devoted to the development of the native iOS app, collaborating with the Web team in the API design, and reinforcing the security of the company through technical and process improvements.



Mobile Engineer Intern (2021 - 2022)

Badger Maps, Inc. — <https://www.badgermapping.com>

Curricular, extracurricular and professional internships at the Californian startup BadgerMaps, Inc. through its engineering branch ZenZorrito Tecnologías S.L. Work on Mobile devoted to code and making decisions for the iOS and Android apps.

Featured projects



DondeTeEsperan (2023 - ...) — <https://dondeteesperan.es>

Online recommendation system for Spanish municipalities based on user preferences and similarity across municipalities. It is the first and only municipality recommender in Spain.

It's powered by a hybrid recommendation system: content-based and collaborative-filter based, and handles 78 variables for the 8,131 Spanish municipalities.

Due to its innovative nature and the problem it solves, it has been awarded by the University of Burgos and Castilla y León Regional Government.

Technologies and techniques used: Python, Pandas, Recommendation Systems, PHP, JavaScript, Flask, Jinja2, Google Cloud, GitHub Actions.

Courses

- iOS foundations for iPhone and iPad - KeepCoding
- Advanced iOS programming - KeepCoding
- Swift 3.0 foundations - KeepCoding
- Artificial Intelligence Foundations. Machine learning and neural networks - University of Salamanca
- Naming strategies: Naming your brand - Crehana
- A technological vision for business people - Udemý
- Android programming from scratch - KeepCoding
- The Complete 2020 Web Development Bootcamp - Dr. Angela Yu, London App Brewery - Udemý
- Complete 2020 Data Science & Machine Learning Bootcamp - Philipp Muellauer, Dr. Angela Yu, London App Brewery - Udemý
- iOS & Swift - The Complete iOS App Development Bootcamp - Dr. Angela Yu, London App Brewery - Udemý
- IV Cybersecurity Conference - Pontifical University of Salamanca
- Python with Django - Telefónica ProFuturo Foundation
- ISO 27001 Internal Auditor - TÜV NORD SPAIN (Cualicontrol)
- ISO 22301 Internal Auditor - TÜV NORD SPAIN (Cualicontrol)
- SwiftUI - Declarative Interfaces for any Apple Device - Mohammad Azam - Udemý

Volunteer experience

- Talk "Optimizing AI for Mobile: Choosing between server vs on-device LLMs An AI-Driven Development (AIDD) case"
 - 2025, Badger Maps, Inc.
- Talk "Taking your code to the next level: Clean Code, SOLID Principles, and Clean Architecture"
 - 2024, ACM USAL Chapter
- Talks "Cybersecurity for minors"
 - 2024, Salesianos Monzón, Secondary - Vocational Training
 - Cybervolunteer at INCIBE
- Help for municipalities affected by DANA
 - November 2024, Volunteer Platform of the Valencian Community
 - Cleaning work carried out in two days (16 hours) in Catarroja and Alfafar
- Private English lessons
 - 2016 - 2017
 - Hermanas Josefinas de la Santísima



BusPlus (2021 - ...) — <https://bmsalamanca.com/empresas>

Comprehensive system for passenger transport companies consisting of native mobile applications and Web platform developed to enable payment by transport card or with NFC compatible and non-NFC compatible mobiles. It has four components: app for end users on Android, app for end users on iOS, app for drivers and points of sale on Android and Web platform for transport companies.

The end users are the passengers, and they can pay without a card, top up their balance, pay with a non-personal card or with a personal card and register their trips to know how much CO2 they save with their trips on public transport.

Drivers receive mobile and card payments on board, and point-of-sale employees update card balances for users who want to top them up with cash.

Regarding management, the transport company's administrative employees control the information displayed to users, review the documentation they send to prove they are in possession of a single-person card, export statistics, and can perform operations at the individual card level.

Technologies and techniques used: NFC, Bluetooth, OTP, Swift, Java, PHP, MySQL, JSON, Firebase, HTML5, CSS3, Javascript, business plan, elevator pitch, copywriting, digital networking & marketing.



BMS - App del Bus Metropolitano de Salamanca (2015 - ...) —

<https://bmsalamanca.com>

First multi-platform app for the public intercity transport of Salamanca. It has three components: Android app, iOS app and online platform used by the managing company *Autobuses Salmantinos* to upload notices about routes and convert them to PDF for its later view by users on the app.

In 2016, started to be supported by *Grupo Avanza-ADO* and became the only official source for accessing, storing and managing the passenger service notices of its subsidiary *Autobuses Salmantinos*.

Since 2018 it is the only way to check the ETAs for each stop. This calculation is provided by an own algorithm which takes into account usual traffic, distance between stops, rush hours or special events, and it is offered through an in-app purchase.

Technologies and techniques used: Objective-C, Java, PHP, MySQL, JSON, Firebase, HTML5, CSS3, Javascript, elevator pitch, copywriting, digital networking & marketing, Google Adwords, Google Cloud Platform.



MemKey - The memes keyboard! (2018 - 2021) —

<https://bmsalamanca.com/memkey>

Mobile application as a custom keyboard for iOS and Android smartphones and tablets that allows users to search a meme through a third-party API and use it or share it on any app.

Technologies and techniques used: Objective-C, Java, REST API, JSON, HTML5, CSS3, Javascript, copywriting, social media, digital networking & marketing.



BurndownChartist (2019 - ...) — <https://burndownchartist.appspot.com/>

Open source web hosted on Google Cloud App Engine and connected to productivity app Todoist's API to produce a *burndown chart* for the specified project, date and options.

Technologies and techniques used: React, REST API, JSON, HTML5, CSS3, Google Cloud, digital networking & marketing.

Trinidad Residencia Sagrada Familia,
Salamanca

- Training and support to achieve the final Senior-High-School level

🏆 Honors and awards

Magna Cum Laude Award at the I Cum Laude Awards for Young Talent, 2025

In recognition of academic and professional achievements, and social commitment. Organized by the University of Salamanca Alumni and the General Foundation of the University of Salamanca.

Extraordinary Master's Degree Award, 2022

Awarded by the University of Burgos, associated with the Official Master's Degree in Business Intelligence and Big Data in Cyber-Secure Environments, completed with an average grade of 3.67/4 GPA, and with a final thesis grade of 3.92/4 GPA.

TCUE Market-Driven Prototypes Contest 2022 Award

One of the six awarded in the Market-Driven Prototypes Contest of the University of Burgos, for the master thesis "Online recommendation system of Spanish municipalities based on user preferences and similarity across municipalities".

2nd Prize - 5th Edition of the Open Data Contest, 2021

Ideas category of the 5th Edition of the Open Data Contest by Castilla y León Regional Government, for the master thesis "Online recommendation system of Spanish municipalities based on user preferences and similarity across municipalities".

TCUE Market-Driven Prototypes Contest 2021 Award

First prize within the Science School and third within the University of Salamanca, including the campuses from Ávila, Béjar and Zamora, on the Market-Driven Prototypes Contest promoted by the University-Business Knowledge Transference Program for the bachelor thesis "Mobile-payment and customer-loyalty system on buses". The prize includes support and training to carry out a business plan following the *lean startup* methodology, and mentoring for the release of a complete and marketable final product.



CartasContraLaHumanidad.es (2019 - ...) —

<https://cartascontralahumanidad.es>

Adaptation from an open source multiplayer online card game, hosted on Heroku and which is the first alternative to play the popular board game *Cards Against Humanity* online in Spanish.

Technologies and techniques used: React, Websockets, Heroic, HTML5, CSS3.



CoViDCyL (2020 - ...) — <https://mariovarona.github.io/CoViDCyL>

Tool to explore online all the data from the set 'Illness rate by basic health zone' over the CoViD-19 incidence on Castile and Leon. It includes a Siri Shortcut to ask her for the current and accumulated incidence at the desired zone. Open source project hosted on Github Pages.

Technologies and techniques used: PHP, API REST, JSON, HTML5, CSS3, Javascript, Github Pages.



Todayist (2020 - ...) — <https://mariovarona.github.io/todayist>

A natural 12-hours-clock to visualize all today tasks from Todoist. Open source project hosted on Github Pages.

Technologies and techniques used: API REST, JSON, HTML5, CSS3, Javascript, Github Pages.

Chronological order



ElPisoDeBilbao.com (2013 - 2014)

Website built from scratch to promote an apartment on sale.

Technologies and techniques used: HTML5, CSS3, responsive web design, Javascript.



Alexia 3.0 (2014)

Operational concept prototype for a possible future version of *Alexia*, the platform for the education sector by *Cospa&Agilmic*. After holding a meeting with its development team, the multinational company included some features from the prototype on the next *Alexia* version.

Technologies and techniques used: HTML5, CSS3, responsive web design, Javascript, elevator pitch, copywriting.



CamerataSalmantina.es (2016 - 2021) — <http://cameratasalmantina.es>

Responsive design web, which offers a great flexibility and speed to keep up to date dynamic content; as galleries or calendars with past and future events, and static content; as information pages or specific *landing pages*.

Technologies and techniques used: HTML5, CSS3, responsive web design, Javascript, PHP, JSON, digital editing.



AngelesCaidosElLibro.com (2017 - 2019)

Backend platform to manage and send an ebook on demand. Frontend web based on WordPress.

Technologies and techniques used: PHP, HTML5, CSS3, WordPress.



Contribution to Open Source Project: Alf.io (2019) — <https://alf.io/>

Translation of almost the entire project from English to Spanish.

Technologies and techniques used: Heroku.

Contribution to Open Source Project: Sweet Alert iOS (2019) —

<https://github.com/mariovarona/SweetAlert-iOS>

Fork from the Sweet Alert iOS project in order to adapt it to Swift 5.

Technologies and techniques used: Swift.



FundHabit (2020 - ...) — <https://mariovarona.github.io/FundHabit>

Tool to incentive and track the habits' achievement through fictional-money contributions to two funds. Open source project hosted on Github Pages.

Technologies and techniques used: HTML5, Cookies, CSS3, Javascript, Github Pages.



Iberdrolux (2020 - ...) — <https://mariovarona.github.io/iberdrolux>

Calculator for *ad hoc* electricity plans, which makes use of previous own hourly consumption data obtained from Iberdrola Distribución Eléctrica (i-de.es). Open source project hosted on Github Pages.

Technologies and techniques used: Python, Github Pages.



MarioVarona (2020 - ...) — <https://mariovarona.dev>

Personal website developed from scratch with a modular approach in which the components are dynamically built based on a JSON file through a Python script. Open source project hosted on Github Pages.

Technologies and techniques used: Python, JSON, HTML5, CSS3, Javascript, Github Pages.



iBloodTests (2025 - ...) — <https://ibloodtests.com>

Artificial Intelligence-based SaaS that analyzes and interprets blood tests and offers dietary, exercise and lifestyle recommendations tailored to the answers entered by users.

Technologies and techniques used: Python, GenAI, Flask, Vue.js, Docker, Google Cloud Platform.